

In 2022, the U.S. National Oceanic and Atmospheric Administration published the average monthly temperatures in the U.S. Census report. The table shows the data for the average recorded temperatures in Jacksonville, where 1 is January.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Highest Temp (°F)	56	67	76	83	88	91	92	91	88	83	76	67

1. Write a quadratic equation, in vertex form, that models the temperature, y , in terms of months, x .

$$y = -(x - 7)^2 + 92$$

2. Write the quadratic equation in standard form.

$$y = -(x^2 - 14x + 49) + 92$$

$$y = -x^2 + 14x + 43$$

3. Use the quadratic equation to determine during what months the average temperature may be 80°F.

$$80 = -(x - 7)^2 + 92$$

$$0 = -x^2 + 14x + 43$$

$$x \approx 3.536 \text{ and } x \approx 10.464$$

March and October

William is playing golf when he hits a shot off the tee. The height of the ball, $h(x)$, in feet, is modeled in the table in terms of x seconds, that the ball has been in the air.

↓ vertex

Seconds, x	1	2	3	4
Height, $h(x)$	80	128	144	128

4. Write a quadratic equation, in vertex form, that models the height of the ball in terms of seconds, x .

$$y = a(x - 3)^2 + 144$$

$$80 = a(1 - 3)^2 + 144$$

$$-64 = 4a$$

$$-16 = a$$

$$y = -16(x - 3)^2 + 144$$

5. Rewrite the quadratic equation modeling William's golf ball's height in standard form.
- $$y = -16(x^2 - 6x + 9) + 144$$
- $$y = -16x^2 + 96x - 144 + 144$$
- $$y = -16x^2 + 96x$$
6. Use the quadratic equation to determine at what time the ball will land back on the ground.
- $$0 = -16x^2 + 96x$$
- $$0 = -16x(x - 6)$$
- $$x = 0, x = 6$$
- After 6 seconds, the ball will land back on the ground.

A mathematician for a music company tracked the sales, in tens of millions of dollars, of music cassette tapes, y , for x years since 1980 up to the year 2000. A table of values of some of the values is shown.

7. Write a quadratic equation, in vertex form, that models the sales from music cassette tapes, y , in terms of years since 1980, x .

$$y = a(x - 9)^2 + 326$$

$$83 = a(0 - 9)^2 + 326$$

$$-243 = 81a$$

$$a = -3$$

$$y = -3(x - 9)^2 + 326$$

x	y
0	83
1	134
3	218
6	299
9	326
10	323
13	278
17	134
19	26

8. Rewrite the quadratic equation in standard form.

$$y = -3(x^2 - 18x + 81) + 326$$

$$y = -3x^2 + 54x - 243 + 326$$

$$y = -3x^2 + 54x + 83$$

vertex →

9. Use the quadratic equation to determine when $y = 0$.

$$a = -3, b = 54, c = 83$$

$$x = \frac{-54 \pm \sqrt{2916 + 996}}{-6}$$

$$x = \frac{-54 \pm \sqrt{3912}}{-6}$$

$$x \approx -1.424, x \approx 19.424$$

10. Interpret the solutions and determine if the solutions are viable in context.

The negative value is not viable it represents 1978 or 1979, which is not part of the year range 1980-2000. In 1999, the sales of cassette tapes reached \$0.